Digital Twinning and LSTM-based Forecasting Model of Solar PV Power Output

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***Abstract*—. The efficiency, return on investment, and output of Solar power plants are all significantly impacted by the development of Digital Twin technology. Digital Twining of the Solar PV system will help to improve the predictability of the power output, gather the necessary information for energy forecasting (predict ahead the power output to balance supply and demand) and power system default detection. The main objective of this study is to develop a Digital Twin Solar PV system. The technical approach used to develop a Solar PV Digital Twin includes: (1) real time data collection of power output from Solar PV system using smart sensors, (2) development, calibration, and validation of a Solar PV system model using MATLAB (3) development and testing the accuracy of LSTM-based forecasting model to predict ahead (15 mins) the Solar PV electrical power output using historical data. The results show that the Solar PV Digital Twin compares well with the targeted data from the real system (data from sensors of the physical Solar PV system). The accuracy of the models was assessed using the coefficient of correlation (R). The results show that the Solar PV Digital Twin simulation and forecasting models have R-values of 0.99893 and 0.99427, respectively. To help an asset owner or operator to visualize the performance of Digital Twin, a dashboard application was developed using MATLAB App Designer. The dashboard shows current, historical, and continuously forecasts future energy consumption and production to balance supply and demand. The development of Digital Twin technology, artificial intelligence and machine learning predictive tools for monitoring, forecasting and maintenance of Solar PV plants are keys to reach the Net Zero Energy Building goals. Additionally, to ensure the reproducibility of the research, the irradiance measurements data was published in a GitHub repository.**

Keywords*—: Solar PV, Digital Twin, Forecasting, LSTM, Net-Zero Energy Building, Modeling & Simulation,

I. INTRODUCTION

To support national net-zero strategies, it's important for building to reduce their energy consumption and offset it with renewable energy (RE). This is critical as one third of global energy goes into buildings [1]. Typically, the first action to reduce building energy consumption is to analyze consumption patterns and implement energy efficiency measures to combat high energy consumption. Once energy consumption is reduced through these measures, the process to investigate RE production potential can begin.

A. Onsite and offsite energy production

1) Onsite vs offsite: Energy production can either be onsite or offsite. Onsite energy production means that energy sources are being utilized at the same location as the load. An example would be Solar heaters installed in households for domestic hot water (DHW) applications. Onsite energy production is more challenging to achieve due to limited space. It gives its users the key advantage of being energy independent. Local production of energy would also the negate the need for energy transmission. Offsite energy production on the other hand is typically handled by a utility company. In this type of energy production, the load is many kilometers or even a city away from the load. Transmission and transportation of the energy is a challenge in the latter case. Ideally, the aim is to achieve both onsite and offsite RE production.

2) Solar PV systems: Solar PV systems have been the preferred onsite energy production system for supplying electricity. Their compact size, ease of installation, aesthetics, and declining prices all contributed to their prevalence. Subsidies have also played a role as certain districts and countries pay certain percentages, fixed amounts, or exempt taxes for homeowners to install Solar PV systems. These systems are usually mounted onto rooftops making roof area the limiting factor in their installation. Hybrid PV-thermal installations are also attractive due to them serving the dual-purpose of supplying electricity and Domestic Hot Water (DHW). They are feasible even in northern countries with proper design [2]. For MENA countries, utility water is already warmed up by the hot climate and further heating is a smaller task compared to colder climates.

B. Net zero energy building

Many new buildings are designed in adherence to new regulations on building energy consumption. Of these regulations, some entice Net Zero Energy Building (NZEB) designs. The term remains vague and open to interpretation. The vagueness comes from defining the energy balance [1]. For most buildings, the energy balance is the balance between electricity consumption and electricity production. This means every kilowatt-hour (KWh) of electricity consumed in the building is also generated by the building. Two main issues in the definition above emerge: timescale of balance and type of energy balance. For the timescale issue, what is the timescale of the balance? Over a day? An hour? A month?

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A year? Over building lifetime? Consumption is highly situational: weekdays and weekends, summer and winter, holidays, etc. Experts and regulators prefer the energy balance for residential buildings to be over a year due to its attainability. Moving onto type of energy balance, we can consider it terms of money, energy, or emissions [3]. By the first type, money, the building must sell to the grid and generate revenue equal to its electricity consumption bill. The target is zero balance at the end of the balancing period. This can be challenging since the price for selling to the grid is usually lower than buying. By the second type, energy, energy sent to the grid has to be equal to energy taken from the grid. The target is 0KWh balance at the end of the balancing period. The third type, emissions, is not a balance exactly since it states that amount of emission-free renewable energy produced should be equal to emission-heavy energy consumed. The target is 0KWh balance at the end of the balancing period. This is easier to obtain compared to the second type since the grid is not entirely fossil-fuel based and we can expect that some energy taken from the grid to be sourced from renewables. For our research, an annual balance of type 2 is the most appealing.

C. *Long short-term memory*

Long Short-Term Memory (LSTM) is a neural network suitable for identifying sequences in the long term but while delivering high short-term results. Previous measurements are used to create the next forecast. Some typical uses of LSTM in literature include forecasting Solar resources [4] or Solar plant production [5]. One benefit of LSTM for Solar energy production is that no weather data such as Solar irradiance or temperature is needed, unlike ANNs.

D. *Digital Twin for energy production*

As the single most important term in our study, Digital Twin is crucial to define. A Digital Twin is a digital copy of a real (physical) system [6]. Digital Twins can forecast future energy production and consumption, provide fault detection, and conduct what-if simulations. Fig. 1 shows an example of a Digital Twin Solar PV plant [7].

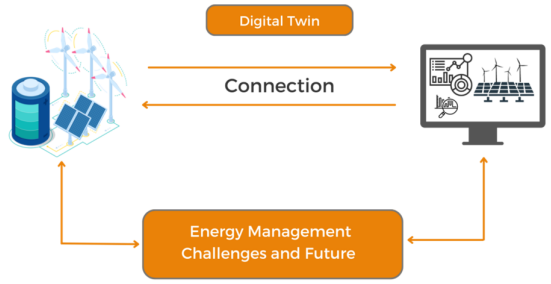


Fig. 1. Digital Twin for Solar PV plants [7]

The main objective of this study is to develop a Digital Twin of Solar PV system. The technical approach used to develop a Solar PV Digital Twin includes: (1) real time data collection of power output from Solar PV system using sensors, (2) developing, calibrating, and validating a Solar PV simulation model using MATLAB (3) developing and testing the accuracy of LSTM-based forecasting model to predict ahead (15 mins) the Solar PV electricity output using historical data.

II. Methodology

A. *Selecting location for Solar PV installation*

The Solar PV system is installed in the roof of Renewable Energy Research laboratory (RE lab), within University of Sharjah premises as shown in Fig. 2 and Fig. 3. The flat rooftop above RE lab is suitable for an onsite Solar PV installation.



Fig. 2. RE lab front view

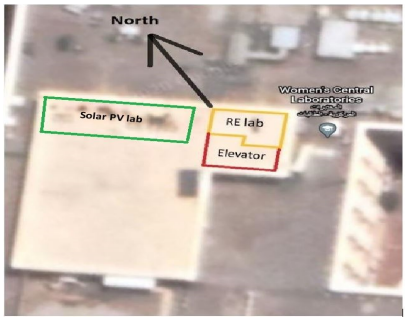


Fig. 3. Google earth image of the RE lab and adjacent rooms

B. *Solar PV production Digital Twin: Simulation model*

In order to create a Digital Twin simulation model for the Solar PV array, generally two things are required: a real system with measurements for inputs and outputs, and a model to obtain simulated output results. These two are then combined to calibrate and validate the Solar PV production model. The validated Solar PV production Digital Twin model can then be used to generate the needed what-if simulations.

1) Solar PV System (Physical System) Power Output Measurement: Two PV panels were purchased from Al-Maden and installed on the top of the RE lab as shown in Fig. 4. These two panels will be used to devlop the digital simulation model. Table I shows the specifications of one of the Solar PV panel. The panels were connected in series and mounted at 25º fixed tilt angle with south orientation. Measurements for maximum power (P_{max}), Global Tilted Irradiance (GTI), and module temperature were taken using a Profitest PV device (see Fig. 5) from sunrise to sunset every 15 minute from the 17th of February 2021 till the 1st of May 2021. The panels and irradiance sensor were cleaned once every two weeks to reduce the effect of soiling on the results. To ensure the reproducibility of the research, the GTI measurements data is published in a GitHub repository [8].

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Fig. 4. Solar PV panels (Physical System)



Fig. 5. Profitest – Solar PV Power Output Measurement Device

TABLE I. SOLAR PV MODULE SPECIFICATIONS

Parameter	Value
Number of cells	72
Module area (m2)	1.956
V_{oc} (V)	45.98
I_{sc} (A)	9.16
V_{mp} (V)	38.25
I_{mp} (A)	8.76
P_{mp} (W)	335
Efficiency (%)	17.1
NOCT (°C)	45
Temperature coefficient of I_{sc} (%/°C)	+0.0484
Temperature coefficient of V_{oc} (%/°C)	-0.2883
Temperature coefficient of P_{max} (%/°C)	-0.3797

2) Solar PV system simulation model: A similar Simulink model that was used by Motahhir [9], was developed and modified in the present study for the simulation model. Fig. 6 illustrates the model developed in this study using Simulink. The model consists of a PV array with irradiance and module temperature as inputs, voltage and current measurement sensors, and a variable DC source. The irradiances and module temperatures were obtained from the PV setup measurements.

The PV array block represents a single diode model (5 parameters model) [10]. The block automatically calculates and extracts the five-parameters for the single diode model from the module data imported by the user. Module data can be modified simply by defining the panel open circuit voltage (V_{oc}), short circuit current (I_{sc}), maximum power point voltage (V_{mp}), maximum power point current (I_{mp}), number of cells, and temperature coefficients for both V_{oc} and I_{sc}, in addition to the numbers of parallel strings and number of modules in series for each string. PV array parameters were tuned to match our panels specifications in Table I.

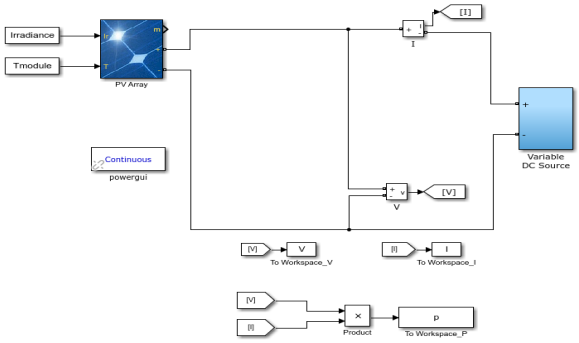


Fig. 6. Solar PV Simulink simulation model

The variable DC source which is shown in Fig. 7 consists of a ramp signal that controls the controlled voltage source. The purpose of using controlled voltage source is to change the PV array output voltage and thus the I-V curve can be extracted for each temperature and irradiance input.

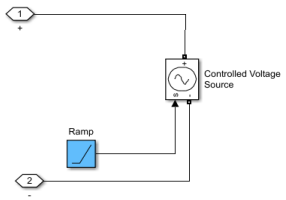


Fig. 7. Variable DC source components

MATLAB and Simulink were used simultaneously to obtain the maximum power and energy of the panels for each irradiance and temperature input. The measured GTI and module temperature from the 17th of February 2021 till the 1st of May 2021 were used to simulate the model and generate the maximum DC power and energy every 15 minutes as outputs. Since the model only simulates the DC power output of the panel, a method for obtaining the useful AC power is needed. System losses other than the losses associated with the electricity generation of the PV panels such as Maximum Power Point Tracking (MPPT) losses, mismatch and wiring losses, and inverter losses have to be taken into consideration. Therefore, a combined efficiency factor of 0.9 was assumed to calculate the AC power produced by multiplying this factor by the DC power.

3) Calibration and validation of the simulation model: An investigation of the Solar PV model results and comparison with the actual measurements were done to evaluate whether the model is accurate enough or needs calibration. Then, a validation of the model was performed by calculating the statistical parameter R from (1) using the actual measurements and simulation results.

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$$R = \sqrt{1 - \frac{\sum_{i=1}^{n}(Actual(i) - simulation(i))^2}{\sum_{i=1}^{n}(Actual(i) - \overline{Actual})^2}} \quad (1)$$

Where Actual is the measured data (real system), Simulation is the results from MATLAB (digital model), and n is the number of data points. $\overline{Actual}$ is the average targeted data (measurement).

C. Solar PV production Digital Twin: Forecasting model

Deep learning techniques have proved their effectiveness in various forecasting applications [11-13]. One of the most accepted methods for forecasting Solar PV production is the usage of Long-Short Term Memory (LSTM) network. Long short-term memory (LSTM) is a type of recurrent neural network. LSTM differs from the more common feedforward neural networks in that it also includes feedback connections. In addition to processing individual data points, such a recurrent neural network is capable of handling complete data sequences. MATLAB was used for developing the LSTM network and evaluating its performance. The LSTM network was used for forecasting the Solar PV electricity output 15 minutes ahead based on historical data. The data from 17th of February 2021 till 24th April 2021 was used in the development of forecasting model. The data was divided into training, testing (forecasting), and validation with approximate percentages of 80%, 10%, and 10%, respectively. Coefficient of Correlation I was calculated using (1) to evaluate the performance of the LSTM network based on the targeted data and network output (predicted data). The rest of the data which was from 25th of April 2021 till 1st of May 2021 was used for further verification of the developed LSTM network forecasting accuracy. Fig. 8 shows the architecture of the LSTM network. It consists of four layers: sequence input layer, LSTM layer, fully connected Layer, and regression layer. The sequence input Layer is based on the number of features, the LSTM layer is based on the number of hidden units, and the fully connected layer is based on the number of responses, which are all shown in table II. The hyperparameters were selected based on a trial-and-error process to maximize the forecasting accuracy. The results for training, testing, validation, combined, and the further verification week are all in AC energy, where the same previous methodology of multiplying the output by 0.9 was implemented.

TABLE II. SELECTED LSTM NETWORK HYPERPARAMETERS

Parameter	Value
Training Algorithm	Adam
Number of Layers	4
Number of Hidden Units	300
Number of Features	1
Number of Responses	1
Number of Epochs	250
Gradient Threshold	1
Initial Learning Rate	0.005
Learn Rate Schedule	piecewise
Learn Rate Drop Period	125
Learn Rate Drop Factor	0.2

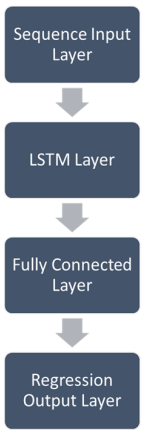


Fig. 8. LSTM Network architecture

III. Results and Discussion

A. Solar PV production Digital Twin: Simulation model

Modeling Solar PV power output is an essential element of the Digital Twin to generate what-if simulations. Solar PV AC power output in 15 minutes timestep results from the simulation model were plotted against the actual measurements (targeted data) in Fig. 9. As seen in the figure, the predictive model had a coefficient of correlation (R) of 0.99893 which indicates a very high accuracy of the model. The Solar PV model has only two inputs (irradiance and temperature), meaning that they are the determinant factors for modeling the Solar PV energy production accurately. Both model inputs are measured onsite in the same Solar PV setup, and the irradiance sensor was cleaned alongside the Solar PV panels once every two weeks. This ensured that all factors that affect the output of the Solar PV setup were captured by the sensors. The figure also shows that all data points. Since the simulation model is validated, it can be used to generate what-if simulations such as simulating the effect of different and extreme operating conditions on the model output.

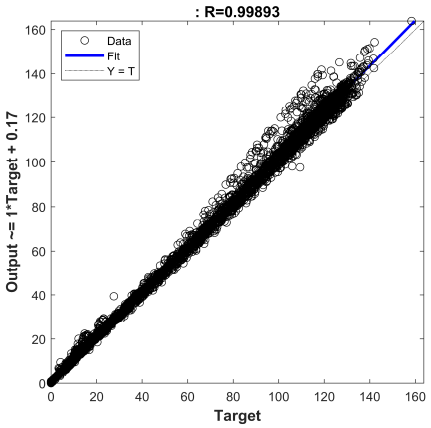


Fig. 9. Coefficient of correlation (R) of the Solar PV model AC energy

B. Solar PV production Digital Twin: Forecasting model

Similar to the building energy consumption (demand) Digital Twin [14], a Solar PV (supply) Digital Twin should have the ability to forecast the future energy production. Fig. 10 shows the training, testing (forecasting), validation, and combined results of LSTM implementation for forecasting the Solar PV AC power output 15 minutes ahead. As shown in the figure, the coefficient of correlation (R) values for training,

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testing, and validation are 0.99413, 0.99831, and 0.99117, respectively. These values indicate that the developed LSTM network has very high accuracy in forecasting the Solar PV energy production. As seen in the figure, most of the data points in all four graphs are near to the diagonal straight line, especially for testing. However, there are few data points, mostly during the training stage, that are far away. This inaccuracy for these data points is most probably the result of rapid variation in weather conditions especially irradiance, caused by fast clouds movement over the Solar PV panels. Another reason for the inaccuracy is that the LSTM network was trained using a small measurement dataset (53 days only) which did not enable the LSTM network to be exposed to extreme and rapidly changing weather conditions.

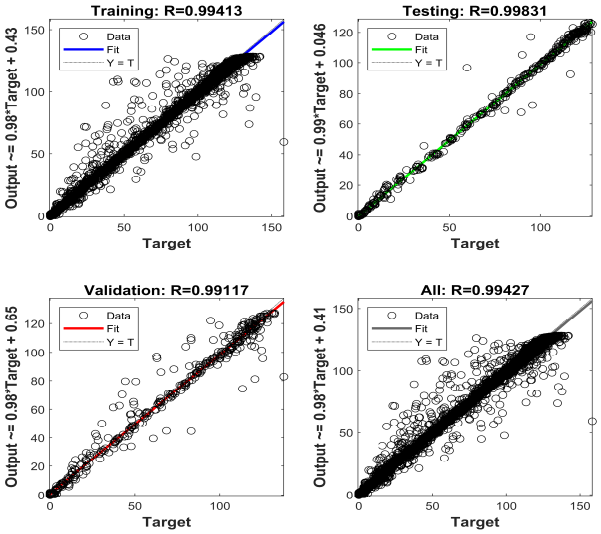


Fig. 10. Coefficient of correlation values of a) Training b) Testing c) Validation d) Combined for the LSTM network

Further verification of the developed LSTM network accuracy was done by comparing the 15 minutes ahead Solar PV AC energy forecasting with the actual measurements for one week. Fig. 11 depicts the comparison between the targeted data (measurement from physical Solar PV system) and the predicted data (Solar PV forecasting model) in 15 minutes timestep. As illustrated in Fig. 11, the forecasting results not only matches the behavior of the actual measurements but also has very high accuracy in terms of the forecasted Solar PV AC energy production values.

C. Balancing consumption and production Digital Twin forecasting models

The goal of designing both demand and supply Digital Twins system forecasting models is to balance between them and explore the opportunity for achieving a Net Zero Energy Building. This can be done by comparing the forecasted values for both Digital Twins (supply and demand). The Digital Twin of the building energy consumption was presented in a previous study by the authors [14], while the Digital Twin of energy production is presented in this paper. Fig. 12 shows the comparison between the actual consumption of RE lab and the potential useful AC energy delivered by our Solar PV setup (with only two panels) in 15 minutes timestep for a week. The figure clearly shows that the two panels cannot supply the enough energy to cover the consumption of RE lab, since the overall Solar PV AC energy is much lower than the total consumption energy. Thus, determining the number of Solar PV panels needed is essential for clear comparison between supply and demand. To illustrate the process of balancing the supply and demand, a semi-sizing process was used to determine the number of panels needed by dividing the total energy consumption by the total Solar PV AC energy production during the sample week. The result of this division is around 7.1. If this number is rounded down to seven, there will be a shortage in energy at the end of the week, and if this number is rounded up to eight, there will be a surplus energy at the end of the week. The number of panels of both cases will be used in the forecasted results during the same sample week. The whole semi-sizing procedure was developed just to mimic an actual shortage or surplus energy case. This can be completely skipped in an actual case scenario where the building owner or operator has a Solar PV system.

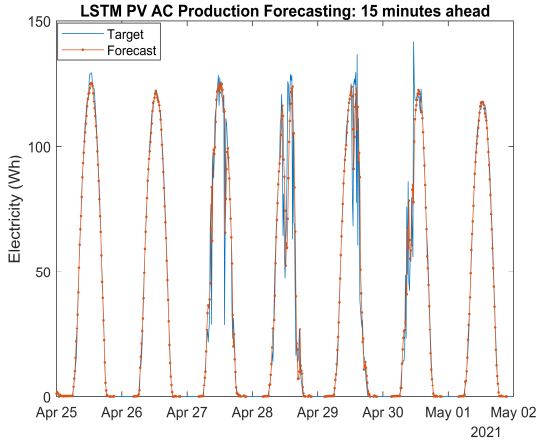


Fig. 11. Solar PV Energy Production: Targeted (meausrement data) versus LSTM-based forecasting data 15 minutes ahead

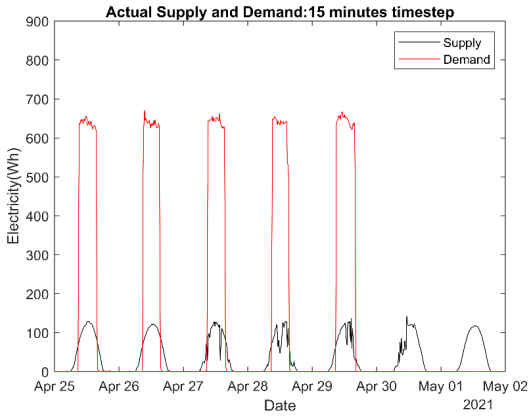


Fig. 12. Actual supply and demand: 15 minutes timestep

To apply the concept of semi-sizing practically, the forecasted Solar PV AC energy for two panels during the sample week was scaled up to the forecasted Solar PV AC energy for the shortage (7 panels) and surplus (8 panels) scenarios, as shown in Fig. 13. This was done by multiplying the forecasting results of 2 panels by 3.5 for the shortage case, and 4 for the surplus case.

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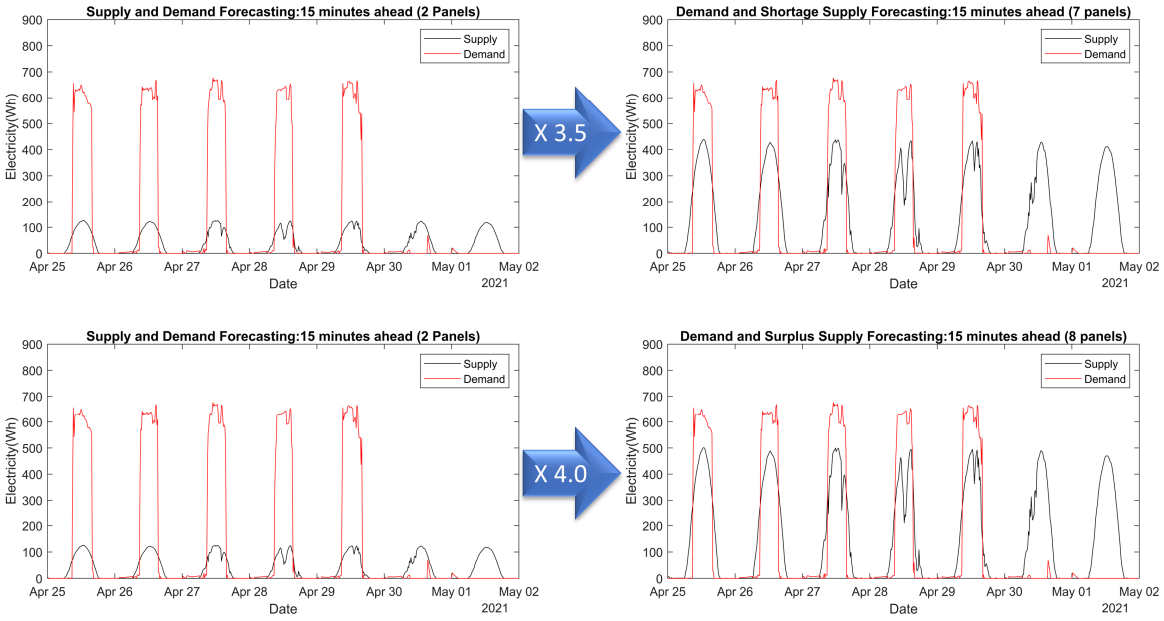


Fig. 13. Supply and demand forecasting 15 minutes timeptep

The top and bottom left hand side of Fig. 13 shows the 15 minutes ahead forecasted RE lab energy consumption, on top of 15 minutes ahead forecasted Solar PV AC energy production for two panels during the sample week. The right-hand side of the figure shows the scaled 15 minutes ahead forecasted Solar PV AC energy for both shortage (top right) and surplus (bottom right) scenarios, on top of the regular 15 minutes ahead forecasted RE lab energy consumption. These graphs on the right-hand side of the figure are helpful to observe the comparison between the supply and demand forecasting visually.

Visualizing the profiles is not adequate to examine the possibility of achieving NZEB for the RE lab. Thus, a new variable called Forecasted Cumulative Net Energy (FCNE) was created. This variable subtracts each 15 minutes ahead forecasted Solar PV AC energy value from each 15 minutes ahead forecasted RE lab consumption value and sums the values up over the sample week. Fig. 14 shows the FCNE values for both shortage (7 panels) and surplus (8 panels) scenarios over the sample week. The variable starts at zero for both scenarios assuming that the Solar PV system has just been installed and the demand prior to this point has been fully covered by the grid. Any rise in the graph indicates that the forecasted supply is higher than the forecasted demand, while any decline in the graph indicates that the forecasted supply is lower than the forecasted demand. Plateaus indicate that the forecasted supply is equal to the forecasted demand. As seen in the figure, the graph starts as a plateau, rises a little afterwards, decreases, increases again in the middle, and then plateaus towards the end of the day. This pattern is repeated during the working days from the 25th of April (Sunday) till the 29th of April (Thursday). The rises during the working days are either because of the supply beginning after sunrise and before the existence of any occupant inside the room, or because of the supply continuing after the occupants leave the room till the sunset. As for the decline, it happens during the occupied hours, where the forecasted supply is always lower than the forecasted demand for both scenarios. Regarding the plateaus during the sample week, they occur before sunrise and after sunset (nighttime), where both forecasted supply and demand are zeroes. One can notice that the graph keeps rising during the weekend (30th of April and 1st May) daytime and flattens during the weekend nighttime. The rises are because of the supply existence during the daytime while the room is not occupied, and the plateaus, as mentioned earlier, occur during the nighttime. One last thing to pay attention to is the two highlighted points at the end of the sample week on the right-hand side of Fig. 14. The highlighted FCNE values for shortage and surplus scenarios are -42.33 Wh and 11952.1 Wh, respectively. The negative value indicates a shortage in the total energy, and the need for an auxiliary electricity source to cover the demand such as the electrical grid or energy storage system. The positive value indicates a surplus total energy that can be either sold to the grid or stored in an energy storage system for later use.

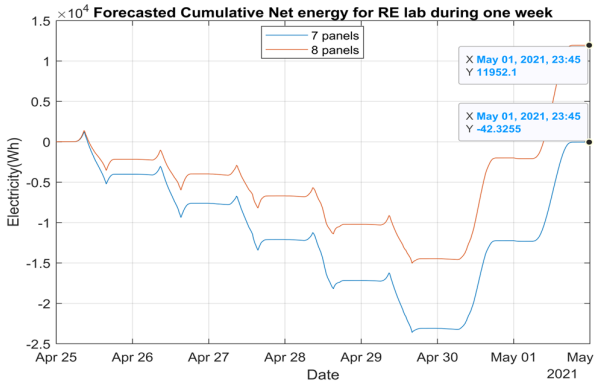


Fig. 14. Forecasted Cumulative Net Energy for RE lab during the sample week

The combination of both consumption and Solar PV production Digital Twins forecasting can continuously and automatically update the FCNE value during operation as shown in Fig 14. This parameter value is highly beneficial for buildings' owners and operators since it offers a quantitative

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evaluation of the net energy of a building. For example, if this value at any given time is negative, it means that there is shortage and the interference of the electrical grid, energy storage system, or an additional onsite renewable energy generation is a must for achieving NZEB. On the other hand, if this value at any given time is positive, it means that there is surplus energy that can be either sold to the grid or stored in an energy storage system. Many different factors determine how to deal with the shortage or surplus energy. One factor is that the grid electricity purchase/selling pricing varies depending on the low/peak demand hours. Another factor is the availability of a small-scale onsite energy storage system, and whether this system can store energy for a long period or not. This is helpful for storing electrical energy during the weekend or holidays and supplying the energy back when needed in commercial buildings just like the case of RE lab. To sum up, at any given point of time and based on the Digital Twins forecasted supply and demand, building's owner/operator or a controller can make proactive decisions on whether the electrical energy should be stored in or taken from an energy storage system, or whether the electrical energy should be purchased from or sold to the grid depending on the electricity pricing and low/peak demand hours.

D. Digital Twins Technologies for Balancing the Supply (Solar PV) and Demnd (Building Energy Conumsption)

A dashboard simplifies access to the Digital Twin for the asset owner or operator. Building upon our initial dashboard prepared using MATLAB App Designer [14], additional visual elements such as gauges and graphs were added for Solar PV energy production. The dashboard would read the energy production measurements and based on them create 15-minutes-ahead forecasts using LSTM network. As shown in Fig. 15, a second tab on Solar PV was added to the dashboard developed for the first part on building energy consumption. This tab starts by showing the current module temperature and irradiance on gauges, followed by a plot displaying produced energy over the last 24 hours in addition to the LSTM network forecasts. Finally, 4 boxes show the forecasted values and live (current) measurements of energy consumption and production.

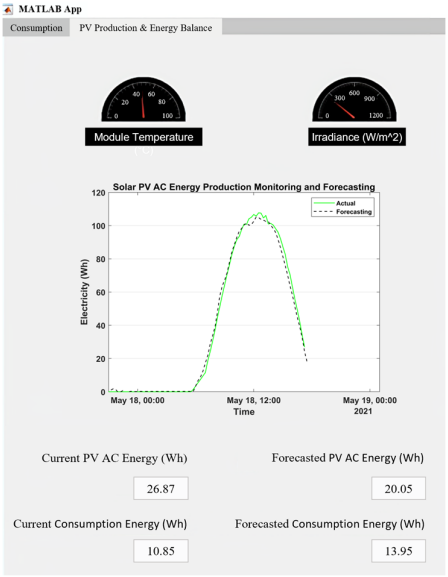


Fig. 15. Digital Twin dashboard showing the supply and demand balance

IV. Conclusion

A new approach for creating a Digital Twin for Solar PV energy production is demonstrated in this study with the goal of balancing energy supply (Solar PV) and demand (building energy consumption). The real Solar PV physical system is equipped with real time data acquisition system for real time measurements of Solar irradiance, Solar module temperature, and power output every 15 minutes. MATLAB/Simulink was used to create and validate a Solar PV simulation model. The Solar PV simulation model results showed very high accuracy (R=0.9989). LSTM network was employed and trained in MATLAB to develop the Digital Twin forecasting model of Solar PV energy production 15 minutes ahead. The R-value for the LSTM network testing (forecasting) was 0.99831, which indicates the reliability of the energy production forecasting from the developed LSTM network. A balance between the consumption and production Digital Twins were done by introducing the "Forecasted Cumulative Net Energy" variable. This variable compares the forecasted consumption with the forecasted production and thus helps buildings' owners/operators or controllers to make proactive decisions about how to manage a shortage/surplus of energy which eases the way towards a Net Zero Energy Buildings. To help an asset owner or operator to visualize the performance of Digital Twin, a dashboard application was developed using MATLAB App Designer. The dashboard shows current, historical, and continuously forecasts future energy consumption and production. This will allow an asset operator to see the balance between consumption and production and forecast future imbalance. Digitalization, monitoring, and forecasting are keys to decarbonization of the energy sector and to reach the NZEB goals. Additionally, to ensure the reproducibility of the research, the irradiance measurements data was published in a GitHub repository.

Future works will include: (1) Deployment of Digital Twin on cloud instead of running it locally, (2) an energy storage system could be incorporated to the energy production Digital Twin, and (3) the Digital Twin's feature set could be expanded to include automated preventative maintenance, and optimization of energy usage based on time-of-use tariffs.

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